MAKOTO M. KELP

L.S. Skaggs Applied Science Building 3279, University of Utah, Salt Lake City, UT 84112
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EDUCATION

Harvard University 2018 – 2023
Ph.D., Earth and Planetary Sciences
S.M., Environmental Science and Engineering
Reed College 2012 – 2016
B.A., Chemistry

PROFESSIONAL APPOINTMENTS

Assistant Professor, University of Utah 2026 – Present
Department of Atmospheric Sciences
Faculty Fellow, Wilkes Center for Climate Science & Policy 2026 – Present
Faculty Fellow, One-U Responsible AI Initiative 2027 – Present
University of Utah
NOAA Climate & Global Change Postdoctoral Fellow, Stanford University 2023 – 2025
Host: Noah Diffenbaugh, with Marshall Burke
Graduate Research Assistant, Harvard University 2018 – 2023
Advisors: Daniel Jacob, Loretta Mickley
Junior Research Scientist, University of Washington 2016 – 2018
Advisor: Julian Marshall

RESEARCH INTERESTS

My research centers on applying machine learning and data-driven methods to uncover new perspectives in atmospheric chemistry and human-environmental systems. I place a special emphasis on exploring the interplay among air pollution, fires, and society.

HONORS AND AWARDS

One-U Responsible AI Initiative Faculty Fellowship 2026
GeoCAFE Scholar (NSF Network on Climate Change and Health for Early Career Scientists) 2026
Utah President's Nominee, Fellowship for Science and Engineering, David and Lucile Packard Foundation 2026
NeurIPS AI4Science Dataset Proposal Competition Winner 2025
NOAA Climate & Global Change Postdoctoral Fellowship 2023 – 2025
Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XVII) 2023
Harvard University Bok Center Certificate of Distinction in Teaching 2019, 2022
AGU Outstanding Student Presentation Award 2019
National Science Foundation STEM Scholar 2013 – 2016

PUBLICATIONS (*SUBMITTED, ADVISEE, †EQUAL CONTRIBUTION)

h-index: 20 (Google Scholar), Published first-authored (9), co-authored (15), and first-author mentored (1) papers:

*32. †Weber, J., and †**M. Kelp**. Atmospheric Chemistry-Climate Modeling with an AI Foundation Model. *Submitted*

*31. Shtein, A., B. Vu, E, Castro, I Kloog, M. Dorman, Y. Wei, L. J. Mickley, **M. Kelp**, and J. Schwartz. Predicting daily PM2.5 concentration for the contiguous United States using an ensemble modeling approach. *Submitted*

*30. Miao, S., R Jin, M. Qiu, **M. Kelp**, D. Shindell, and L. He. Unequal burden of compound heat–smoke extremes across the United States. *Submitted*

*29. Hu, J. Y., I. Higuera-Mendieta, P. O. Sturm, and **M. Kelp**. Toward Mechanistic Interpretability of an AI Foundation Model Fine-Tuned for Atmospheric Chemistry. *Submitted*

*28. †**Kelp, M.**, †M. Qiu, S. Heft-Neal, R. Sandoval, C. Schollaert, M. Marlier, M. Burke, and N. S. Diffenbaugh. Competing Health Impacts of Smoke from Wildfires, Agricultural Burning, and Prescribed Fires in California. *Submitted*

*27. †Zhu, Q., †T. Liu, and †**M. Kelp**. Capturing Time-Resolved Prescribed Fire Emissions with TEMPO Special Observations. *Submitted*

*26. Qiu, M., Y. Li, M. Childs, **M. Kelp**, X. Jin, G. Huang, M.D. Yazdi, Y. Wei, Y. Wang, and K. Chen. Wildfire smoke offsets decades of progress in reducing ozone exposure across the United States. *Submitted*

25. Liu, T., K. Otobe, and **M. Kelp** (2026). Tracking the efficacy of prescribed burns in three phases: fuel removal, wildfire mitigation, and vegetation recovery. *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, XLIX-B3-2026, 987-993, DOI: 10.5194/isprs-archives-XLIX-B3-2026-987-2026

24. Li, Y., X. Jin, **M. Kelp**, H.Z. Sun, and M. Qiu (2026). Growing impacts of fire smoke on ozone pollution and associated mortality burden in the United States. *Science Advances*, 12, 18, DOI:10.1126/sciadv.aec2903.

23. Feng, X., L.J. Mickley, J.O. Kaplan, **M. Kelp**, Y. Li, and T. Liu (2025). Large role of anthropogenic climate change in driving smoke concentrations across the western United States from 1992 to 2020. *Proc. Natl. Acad. Sci.*, 122 (49) e2421903122, DOI: 10.1073/pnas.2421903122.

22. †Hickman, S., †**M. Kelp**, †P. Griffiths, †K. Doerksen, †K. Miyazaki, †E. Pennington, †G. Koren, †F. Iglesias-Suarez, †M. Schultz, K. Chang, O.R. Cooper, A. Archibald, R. Sommariva, D. Carlson, H. Wang, J. West, and Z. Liu (2025). Applications of Machine Learning and Artificial Intelligence in Tropospheric Ozone Research. *Geosci. Model Dev.*, 18, 8777–8800, DOI: 10.5194/gmd-18-8777-2025.
Selected as Highlight Paper
Tropospheric Ozone Assessment Report Phase II (TOAR-II) Community Special Issue

21. Qiu, M., J. Li, C.F. Gould, R. Jing, **M. Kelp**, M.L. Childs, J. Wen, Y. Xie, M. Lin, M.V. Kiang, S. Heft-Neal, N.S. Diffenbaugh, and M. Burke (2025). Wildfire smoke exposure and mortality burden in the US under future climate change. *Nature*, 647, 935–943, DOI: 10.1038/s41586-025-09611-w.

20. **Kelp, M.**, M. Burke, M. Qiu, I. Higuera-Mendieta, T. Liu, and N. Diffenbaugh (2025). Effect of Recent Prescribed Burning and Land Management on Wildfire Burn Severity and Smoke Emissions in the Western United States. *AGU Advances*, 6, e2025AV001682, DOI: 10.1029/2025AV001682.
Editor Pick from AGU Advances

19. Chung, K., T. Liu, **M. Kelp**, K. Vohra, D. Skelly, M. Carroll, J. Schwartz, and L.J. Mickley (2025). Managing Smoke Risk from Wildland Fires: Northern California as a Case Study. *Environ. Sci. Technol.*, 59, 27, 13912–13923, DOI: 10.1021/acs.est.5c01914.
Advised Undergraduate First-Author with T. Liu

18. Qiu, M., D. Chen, **M. Kelp**, J. Li, G. Huang, M.D. Yazdi (2025). The rising threats of wildland-urban interface fires in the era of climate change: The Los Angeles 2025 fires. *The Innovation*, Volume 6, Issue 5, DOI: 10.1016/j.xinn.2025.100835.

17. Kawano, A., **M. Kelp**, M. Qiu, K. Singh, E. Chaturvedi, I. Azevedo, and M. Burke (2025). Improved daily $PM_{2.5}$ estimates in India reveal inequalities in recent enhancement of air quality. *Science Advances*, 11, 4, DOI: 10.1126/sciadv.adq1071.

16. Qiu, M., **M. Kelp**, S. Heft-Neal, X. Jin, C.F. Gould, D.Q. Tong, and M. Burke (2024). Evaluating Estimation Methods for Wildfire Smoke and their Implications for Assessing Health Effects. *Environ. Sci. Technol.*, 58, 52, 22880–22893, DOI: 10.1021/acs.est.4c05922.
Special Issue on "Wildland Fires: Emissions, Chemistry, Contamination, Climate, and Human Health"

15. Liu, T., F.M. Panday, M.C. Caine, **M. Kelp**, D.C. Pendergrass, and L.J. Mickley (2024). Is the smoke aloft? Caveats regarding the use of the Hazard Mapping System (HMS) smoke product as a proxy for surface smoke presence across the United States. *International Journal of Wildland Fire*, 33, WF23148, DOI: 10.1071/WF23148.

14. Lin, H., L.K. Emmons, E.W. Lundgren, L.H. Yang, X. Feng, R. Dang, S. Zhai, Y. Tang, **M. Kelp**, N.K. Colombi, S.D. Eastham, T.M. Fritz, A.M. Fiore, and D.J. Jacob (2024). Intercomparison of GEOS-Chem and

CAM-chem tropospheric oxidant chemistry within the Community Earth System Model version 2 (CESM2). *Atmospheric Chemistry and Physics*, 24, 8607–8624, DOI: 10.5194/acp-24-8607-2024.

13. **Kelp, M.**, C. A. Keller, K. Wargan, B.M. Karpowicz, and D. J. Jacob (2023). Tropospheric ozone data assimilation in the NASA GEOS Composition Forecast modeling system (GEOS-CF v2.0) using satellite data for ozone vertical profiles (MLS), total ozone columns (OMI), and thermal infrared radiances (AIRS, IASI). *Environ. Res. Lett.*, 18, 094036, DOI: 10.1088/1748-9326/acf0b7.

12. **Kelp, M.**, T. C. Fargiano, S. Lin, T. Liu, J.R. Turner, J. N. Kutz, and L.J. Mickley (2023). Data-driven placement of $PM_{2.5}$ air quality sensors in the United States: an approach to target urban environmental injustice, *GeoHealth*, 7, e2023GH000834, DOI: 10.1029/2023GH000834.
Special Collection on "Geospatial data applications for environmental justice"

11. Balasus, N., D. J. Jacob, A. Lorente, J. D. Maasakkers, R. J. Parker, H. Boesch, Z. Chen, **M. Kelp**, H. Nesser, and D. J. Varon (2023). A blended TROPOMI+GOSAT satellite data product for atmospheric methane using machine learning to correct retrieval biases. *Atmos. Meas. Tech.*, 16, 3787–3807, DOI: 10.5194/amt-16-3787-2023.

10. **Kelp, M.**, M. Carroll, T. Liu, R.M. Yantosca, H.E. Hockenberry, and L.J. Mickley (2023). Prescribed burns as a tool to mitigate future wildfire smoke exposure: Lessons for states and environmental justice communities. *Earth's Future*, 11, e2022EF003468, DOI: 10.1029/2022EF003468.
Among *Earth's Future* Top Viewed Articles of 2023

9. **Kelp, M.**, D.J. Jacob, H. Lin, and M.P. Sulprizio (2022). An online-learned neural network chemical solver for stable long-term global simulations of atmospheric chemistry. *JAMES*, 14, e2021MS002926, DOI: 10.1029/2021MS002926.
Selected as Highlight Paper
Special Collection on "Machine learning application to Earth system modeling"

8. Yang, L. H., D.H. Hagan, J.C. Rivera-Rios, **M. Kelp**, E.S. Cross, C.Y. Peng, J. Kaiser, L.R. Williams, P. L. Croteau, J.T. Jayne, and N.L. Ng (2022). Investigating the sources of urban air pollution using low-cost air quality sensors at an urban Atlanta site. *Environ. Sci. Technol.*, 56, 11, 7063–7073, DOI: 10.1021/acs.est.1c07005.

7. **Kelp, M.**, S. Lin, J.N. Kutz, and L.J. Mickley (2022). A new approach for optimal placement of $PM_{2.5}$ air quality sensors: case study for the contiguous United States. *Environ. Res. Lett.*, 17, 034034, DOI: 10.1088/1748-9326/ac548f.

6. **Kelp, M.**, D.J. Jacob, J.N. Kutz, J.D. Marshall, and C.Tessum (2020). Toward stable, general machine-learned models of the atmospheric chemical system. *JGR: Atmospheres*, 125, e2020JD032759, DOI: 10.1029/2020JD032759.

5. **Kelp, M.**, T. Gould, E. Austin, J.D. Marshall, M. Yost, C. Simpson, and T. Larson (2020). Sensitivity analysis of area-wide, mobile source emission factors to high-emitter vehicles in Los Angeles. *Atmospheric Environment*, 223, 117212, DOI: 10.1016/j.atmosenv.2019.117212.

4. Wen, Y., H. Wang, T. Larson, **M. Kelp**, S. Zhang, Y. Wu, and J.D. Marshall (2019). On-highway vehicle emission factors, and spatial patterns, based on mobile monitoring and absolute principal component score. *Science of The Total Environment*, 676, 242-251, DOI: 10.1016/j.scitotenv.2019.04.185.

3. **Kelp, M.**, A.P. Grieshop, C.O. Reynolds, J. Baumgartner, G. Jain, K. Sethuramanand, and J.D. Marshall (2018). Real-time indoor measurement of health and climate-relevant air pollution concentrations during a carbon-finance-approved cookstove intervention in rural India. *Development Engineering*, 3, 125-132, DOI: 10.1016/j.deveng.2018.05.001.

2. Brewer, J. F., M. Bishop, **M. Kelp**, C. Keller, A.R. Ravishankara, and E.V. Fischer (2017). A sensitivity analysis of key factors in the modeled global acetone budget. *J. Geophys. Res.*, 122, DOI: 10.1002/2016JD025935.

1. Jaffe, D., J. Putz, G. Hof, G. Hof, J. Hee, D.A. Lommers-Johnson, F. Gabela, J. Fry, B. Ayres, **M. Kelp**, and M. Minsk (2015). Diesel particulate matter and coal dust from trains in the Columbia River Gorge, Washington state, USA. *Atmospheric Pollution Research*, 6, 946-952, DOI: 10.1016/j.apr.2015.04.004.

WORKSHOP AND WHITE PAPERS

4. **Kelp, M.**, A. Arzani, P. Castellanos, P. Griffiths, I. Higuera-Mendieta, M. Perez-Carrasco, V. Shah, P.O. Sturm, J. Weber (2026). Strategic Governance of AI Models in Earth Science. *arXiv*:2610.10560.
3. K. Miyazaki, J. Barré, M. Abdi-Oskouei, A. Arellano, K. Bowman, B. Gaubert, B. Handerson, C.-H. Hsu, **M. Kelp**, B. McDonald, D. Millet, A. Mizzi, D. Pendergrass, A. Raman, V. Shah, T. Stavrakou, Y. Wang, K. Wargan, B. Weir, L. Wu, J. Yu (2026). Understanding Atmospheric Change: Integrating Observations and Models for Emissions, Prediction, and Future Missions. (Submitted)
2. **Kelp, M.**, S. Hickman, K. Miyazaki, K. Chang, P. Griffiths, Q. Zhu, G. Koren, F. Iglesias-Suarez, E. Pennington, and M.G. Schultz (2025). AI4O3: A Foundational Data Collection for Artificial Intelligence in Tropospheric Ozone Research. *NeurIPS 2025 Workshop on AI4Science* (Oral, Dataset Track), DOI: openreview.net/forum?id=JuEZ5F8E3
   Dataset Proposal Competition Award Winner
1. **Kelp, M.**, C. Tessum, and J.D. Marshall (2018). Orders-of-magnitude speedup in atmospheric chemistry modeling through neural network-based emulation. *arXiv*:1808.03874.

INVITED TALKS AND SEMINARS (*UPCOMING)

*2026 Carnegie Mellon University – Department of Chemical Engineering Seminar Series
*2026 Stony Brook University – School of Marine and Atmospheric Sciences Seminar Series
2026 NASA JPL – Atmospheric Composition Assimilation and Modeling Seminar Series
2026 University of Reading – Meteorology
2026 NASA GMAO Seminar Series
2026 AGU/AMS Climate & Health Showcase
2026 American Thoracic Society Workshop
2026 University of Toronto – Noble Seminar Series
2026 Utah Center for Data Science
2026 Stanford Woods Institute – Wildfire Working Group
2025 Cascadia Wildfire & Urban Smoke Webinar
2025 CPAESS Discovery Series Seminar
2025 UC Berkeley – CEE Environmental Engineering Seminar
2024 UC Irvine – Earth System Science
2024 NOAA Climate and Global Change Summer Institute
2024 MIT – Civil and Environmental Engineering
2024 University of Washington – Atmospheric Science and Computer Science
2024 University of Utah – Atmospheric Science
2024 AI for the Study of Environmental Risk (AI4ER) seminar, University of Cambridge
2023 Southwest Fire Science Consortium and Fire Adapted New Mexico Learning Network
2023 PSEG Institute for Sustainability Studies with The Trust for Public Land, Montclair State University
2023 Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS) XVII, Brookhaven National Lab
2023 NASA GISS
2023 Columbia University
2023 Science in the News, Harvard University
2023 AGU/AMS GeoHealth Showcase
2023 MIT Atmospheric Chemistry Colloquium
2023 Stanford University
2022 Royal Meteorological Society Atmospheric Chemistry Special Interest Conference
2022 Pennsylvania Department of Environmental Protection Air Monitoring Committee Workshop

2022 Karlsruhe Institute of Technology

2022 University of Washington

2022 University of Illinois at Urbana-Champaign Advanced Environmental Engineering Seminar

2022 EPA Model Applications Team Meeting

SELECTED CONFERENCE PRESENTATIONS (†INVITED, *UPCOMING)

*†2027 Gordon Research Conference in Atmospheric Chemistry, Newry, ME (Discussion Leader)

*†2027 Uniting People for Livable Intermountain Futures Together (UPLIFT) NSF Workshop, Tuscon, AZ (Talk)

*2026 AGU Fall Meeting, San Francisco, CA (Talk x2)

*†2026 AI2 Climate Modeling Conference: "What Role Does Physics Play in Data-Driven Climate Modeling?", Seattle, WA (Talk)

2026 Deep Learning for Science (DL4Sci), Lawrence Berkeley National Laboratory, Berkeley, CA (Poster)

2025 AGU Fall Meeting, New Orleans, LA (Talk)

2025 NeurIPS 2025 Workshop AI4Science, San Diego, CA (Talk)

2025 Gordon Research Conference in Atmospheric Chemistry, Newry, ME (Poster)

2024 AGU Fall Meeting, Washington D.C. (Talk)

†2024 Health Effects Institute Annual Conference, Philadelphia, PA (Talk)

2023 AGU Fall Meeting, San Francisco, CA (Talk)

†2023 Meteorology and Climate - Modeling for Air Quality, Session: "Breakthrough Innovations in Atmospheric & Air Quality Modeling", UC Davis (Talk)

2023 Gordon Research Conference in Atmospheric Chemistry, Newry, ME (Poster)

2023 AMS Annual Meeting, Denver, CO (Talk)

2022 AGU Fall Meeting, Chicago, IL (Talk x2)

†2022 Atmospheric Chemical Mechanisms, Session: "Mechanism Development and Reduction", UC Davis (Talk)

2022 AMS Annual Meeting, Virtual (Talk)

2022 10th International GEOS-Chem Meeting, Washington University in St. Louis (Talk)

†2022 ECMWF Machine Learning Workshop (Talk)

2021 AMS Annual Meeting, Virtual (Talk)

†2020 AGU Fall Meeting, Virtual (Talk)

2020 Atmospheric Chemical Mechanisms Conference, Virtual (Talk)

2019 AGU Fall Meeting, San Francisco, CA (Poster)

2019 Deep Learning for Science (DL4Sci), Lawrence Berkeley National Laboratory, Berkeley, CA (Poster)

2018 AGU Fall Meeting, Washington D.C. (Poster)

TEACHING EXPERIENCE

University of Utah Atmospheric Sciences
ATMOS/BIOL/GEO 1120: Introduction to Earth System Science **Fall 2026**
ATMOS 3100/6090: Atmospheric Chemistry **Spring 2027**
ATMOS 5450/6450: Machine Learning for the Environmental Sciences (course developer) **Spring 2027**

Stanford University Civil and Environmental Engineering
CEE265F/SUSTAIN248: Environmental Governance and Climate Resilience
Guest Lecturer **Winter 2025**

Harvard University Department of Earth and Planetary Sciences
EPS 200: Graduate-level Atmospheric Chemistry and Physics
Teaching Fellow **Fall 2019, Fall 2020, Fall 2022**
Guest Lecturer **Fall 2022**

Derek Bok Center for Education & Learning

Teaching Certificate **2023**

Reed College Department of Chemistry
Chem 101: Molecular Structure and Properties
Chem 102: Chemical Reactivity
Chem 230: Environmental Chemistry

Laboratory Teaching Assistant **2015-2016**

Tutor, Grader **2013-2016**

RESEARCH MENTORING

Graduate Students:

- Liam Thompson (UU ATMOS), Fall 2026-present

Undergraduates:

- Bishwo Jeet Bista (Summer 2026-Present, Harvard University): *"Centralized prescribed fire database for the US"*
- Jason Hu (Winter 2025 – Spring 2026, Stanford University): *"Mechanistic Interpretability of Earth system foundation models for air pollution"*
  - ⋆ Lead author of Hu et al. (submitted); paper accepted at NeurIPS 2026 Workshop XAI4Science
  - ⋆ Oral presentation at Stanford Sustainability Data Science Conference (2026; Undergraduate Rising Star Award)
  - ⋆ Poster presentation at AGU Fall Meeting (2025)
- Howard Qin (Spring 2025-Fall 2025, Stanford University): *"Physics Benchmarking for AI Numerical Weather Prediction Models"*
  - ⋆ Poster presentation at AMS Annual Meeting (2026)
- Nia McKenzie (Summer 2025, Spelman College): *"Developing a comprehensive and centralized database of prescribed fire in the United States"*
- Emily Jones (Summer 2025, Stanford University): *"Tracking Air Pollution from Unregulated Emissions at NASCAR Races and NFL Tailgates"*
- Rebecca Sandoval (Summer 2024, University of California, Santa Barbara): *"Evaluating prescribed fire emissions in California"*
- Lyna Kim (Spring 2024 –Summer 2025, Stanford University): *"Physically interpretable machine learning for climate and weather forecasting"*
  - ⋆ Poster presentation at AGU Fall Meeting (2024)
- Christian Chiu (Summer 2023 – Fall 2023, Harvard University): *"Uncovering latent VOC emissions and spatiotemporal drivers of urban ozone in changing NO_x regimes"*
- Karina Chung (Summer 2023 – Fall 2024, Harvard University): *"Google Earth Engine applications for wildfire smoke risk in the Western United States"*
  - ⋆ Lead author of Chung et al. (2025) *Environ. Sci. Technol.*, undergraduate student feature
  - ⋆ Selected for AGU/AMS Climate and Health Showcase (2024)
  - ⋆ Oral presentation at AGU Fall Meeting (2023)
- Greta Schultz (Summer 2023, University of Wisconsin-Madison): *"Emergency mobile monitoring for California wildfire smoke"*
  - ⋆ Poster presentation at AMS Annual Meeting (2024)
- Timothy Fargiano (Summer 2022 – Fall 2022, Harvard University): *"Optimal placement of $PM_{2.5}$ air quality sensors in the US: An approach to target environmental injustice"*
  - ⋆ Co-author on Kelp et al. (2023b)
- Margaret Schultz (January 2022 – December 2022, Harvard University): *"Real-time high-resolution downscaling of fine particulate matter ($PM_{2.5}$) air quality in the United States using machine learning"*
  - ⋆ Year-long environmental engineering senior thesis: senior project spotlight

- Samuel Lin (Summer 2021- Fall 2021, Harvard University): *"Optimal air quality sensor placement in the United States"*
  - ⋆ Co-author on Kelp et al. (2023b, 2022a)
- Marie Panday (Summer 2021, University of Maryland): *"Trends in and Reconstruction of Smoke Days across the United States"*
  - ⋆ Co-author on Liu et al. (2024)
- Kent Toshima (Summer 2020 - Summer 2021, Harvard University): *"Application of deep learning to detection of wildfire smoke in HMS over North America"*
- Miah Caine (Summer 2020 - Spring 2021, Harvard University): *"Agreement between the HMS Product and Ground-Level Smoke in the Pacific Northwest"*
  - ⋆ Co-author on Liu et al. (2024)

Committees as member:

- Haley Humble (UU ATMOS), PhD Committee 2026-
- Ben Hartvigsen (UBC EOAS), PhD Committee 2026-
- Zack Zounes (UU ATMOS), PhD Committee 2026-
- Corey Bois (UU ATMOS), PhD Committee 2025-
- Vanessa Sun (UU ATMOS), PhD Committee 2024-

SYNERGISTIC ACTIVITIES

International:

Technical AI Safety Certification from BlueDot Impact talent accelerator *(selective cohort program; 2026)*

Selected participant for Deep Learning for Science Summer School, Lawrence Berkeley National Lab *(2026)*

Co-Chair for Tropospheric Ozone Assessment Report, Phase II (TOAR-II) Machine Learning for Tropospheric Ozone (ML4O3) Working Group *(2023 – 2025)*

Co-leader of Statistical Learning in Atmospheric Chemistry (SLAC) group, seminar series (2022 – Present)

Proposal reviewer for *NASA, NSF, Spark Climate Solutions, NSERC (Canada)*

Peer reviewer for *Nature, Science, PNAS, Atmospheric Chemistry and Physics, Geophysical Research Letters, Environmental Research Letters, JAMES, GeoHealth, Environmental Science & Technology, Earth's Future, ES&T Air, Geoscientific Model Development, Nature Ecology & Evolution, Atmospheric Environment, ACS Earth and Space Chemistry, Communications Earth & Environment, Environmental Research Communications, Atmospheric Pollution Research, Environmental Data Science, Environmental Science, and Pollution Research*

National:

Co-Chair for Atmospheric Chemistry Gordon Research Seminar (GRS) *(2025)*

Technical Program Committee for Atmospheric Chemical Mechanisms conference *(2024)*

Session Convener at AGU Fall Meeting:

– Data-Driven Methods for Quantifying Atmospheric Composition: Advances in Computation and Statistical Learning *(2023, 2024, 2025, 2026)*

– Air Pollution and Meteorology Extremes: Interactions, Trends, and Public Health Dimensions *(2025, 2026)*

– Data-Driven Policy for Air Quality and Climate Resilience *(2026)*

– Managing Air Quality and Public Health Under Climate Change: Health Impacts, Mitigation Policy, and Adaptation Strategies *(2024)*

– Prescribed Fires and Land Management in North America *(2023)*